

Cost Analysis

2026-08-10

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1 Assumptions

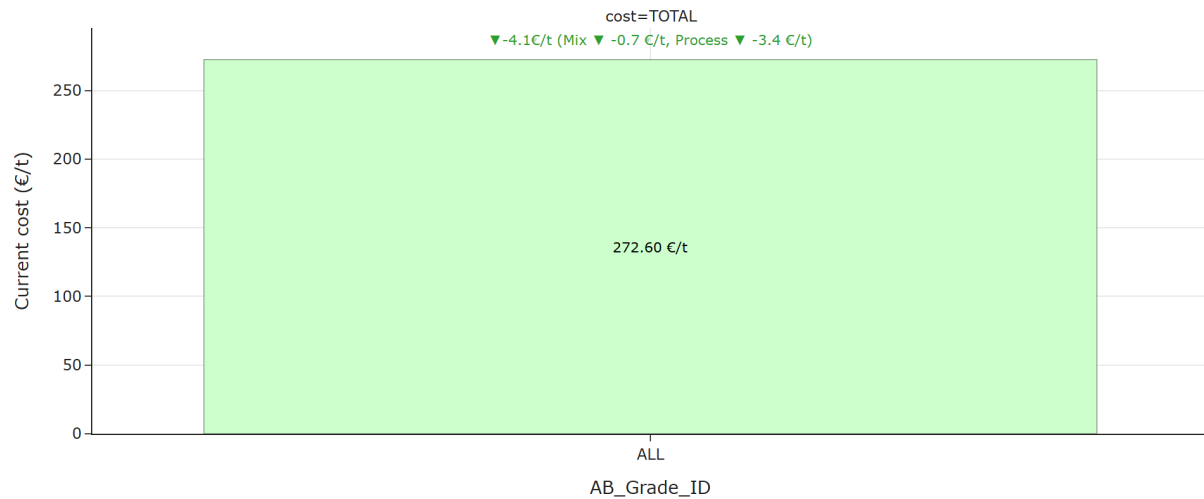
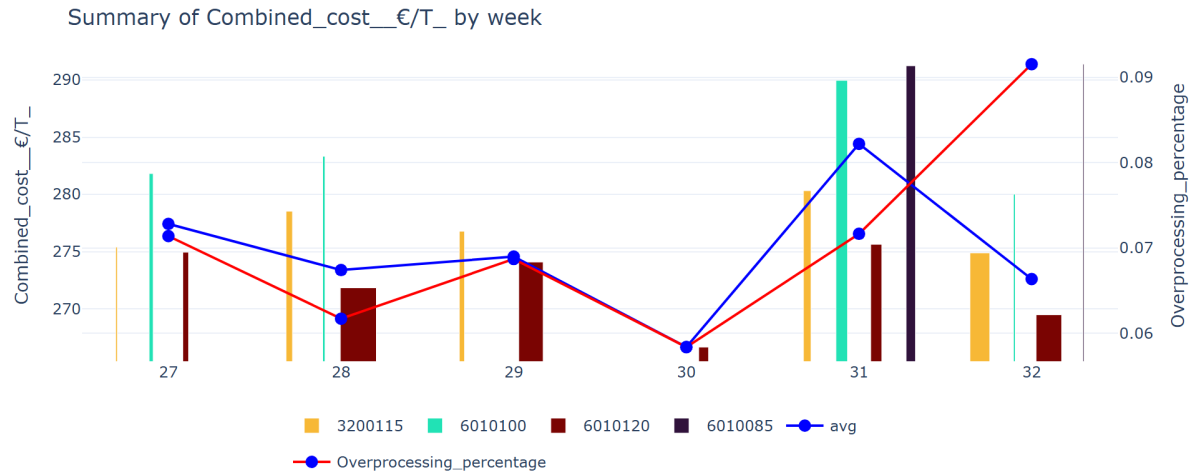
The aim of the analysis is to identify the most cost-effective production periods in months of data and recommend optimal operating windows for the corresponding key influencers that can be set as a benchmark for the weekly analysis.

- **Period Analyzed:** July 05, 2026 – August 10, 2026
- **Cost Components Included:** Fibre, steam, electricity, starch, and chemicals.
- **Grades Included:** Only grades 6010085, 6010100, 601020 and 3200115 are included in this week’s analysis.
- **Data Selection Criteria:** Only operating conditions where strength properties exceed specification thresholds were considered.
- **Naming Conventions Used:**
 - **Current:** Refers to the last week of the analyzed period.
 - **Historic:** Represents the average cost over the monthly interval previous to the “Current” period. Used as baseline.
 - **Best:** Corresponds to the cost associated with the optimal batch configuration.

2 Executive Summary

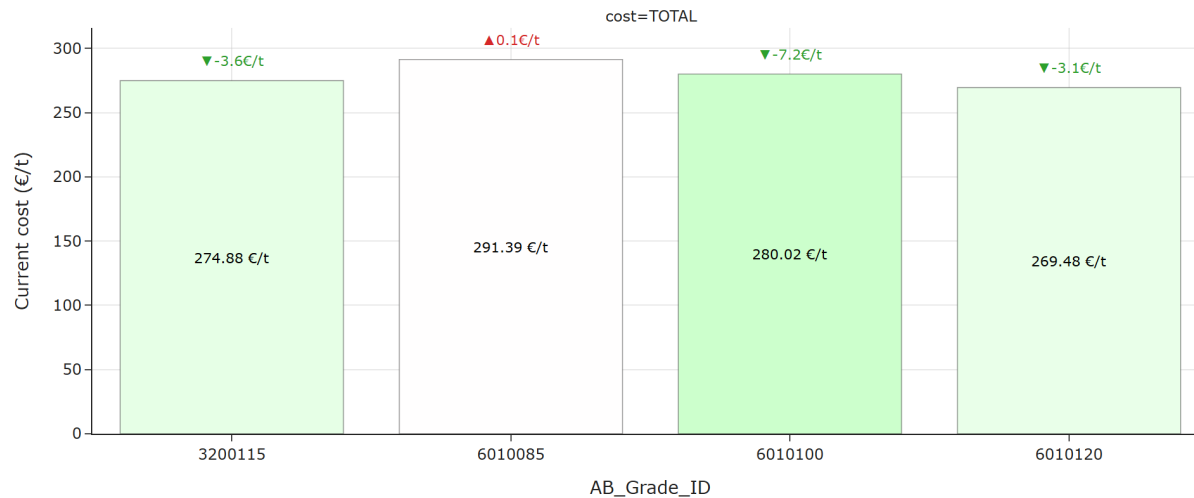
The following plot illustrates how total cost varies across different grades, enabling a comparison between the current configuration and the historical baseline.

The next plot displays how total cost evolves over time, with a breakdown by grade to show individual contributions.



For all considered grades, TOTAL cost saw a decrease of 4.12 €/t, resulting in 272.60 €/t.

Breaking down the overall cost change, -0.75 €/t was driven by a cheaper mix of grades, and -3.37 €/t was due to an improvement in the process.



For grade 3200115, TOTAL cost saw a decrease of 3.60 €/t, resulting in 274.88 €/t.

For grade 6010085, TOTAL cost saw an increase of 0.14 €/t, resulting in 291.39 €/t.

For grade 6010100, TOTAL cost saw a decrease of 7.25 €/t, resulting in 280.02 €/t.

For grade 6010120, TOTAL cost saw a decrease of 3.09 €/t, resulting in 269.48 €/t.

Looking across individual grades, the strongest cost increase was observed for grade 6010085 (+0.14 €/t) and the strongest cost decrease was observed for grade 6010100 (-7.25 €/t). These movements stand out and may require further investigation.



Average change by component:

- **Steam_€/T** decreased on average (-3.39 €/t).
- **Starch_€/T** decreased on average (-0.88 €/t).
- **Electricity_€/T** increased on average (+0.52 €/t).
- **Fibre_cost_€/T** increased on average (+0.23 €/t).
- **Chemicals_€/T** increased on average (+0.07 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Electricity_€/T** (+0.52 €/t on average), with the highest increase in grade 6010100 (+1.41 €/t) and the largest average decrease comes from **Steam_€/T** (-3.39 €/t on average), with the biggest drop in grade 6010100 (-6.92 €/t). These components stand out and may require further investigation.



Average change by steam:

- **Steam_Predryers** decreased on average (-3.15 €/t).
- **Steam_Whitewater** decreased on average (-0.64 €/t).
- **Steam_Afterdryers** increased on average (+0.40 €/t).
- **Steam_Starchkitchen** increased on average (+0.00 €/t).
- **Steam_Heatexchangers** increased on average (+0.00 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Steam_Afterdryers** (+0.40 €/t on average), with the highest increase in grade 3200115 (+0.99 €/t) and the largest average decrease comes from **Steam_Predryers** (-3.15 €/t on average), with the biggest drop in grade 6010100 (-5.57 €/t). These components stand out and may require further investigation.

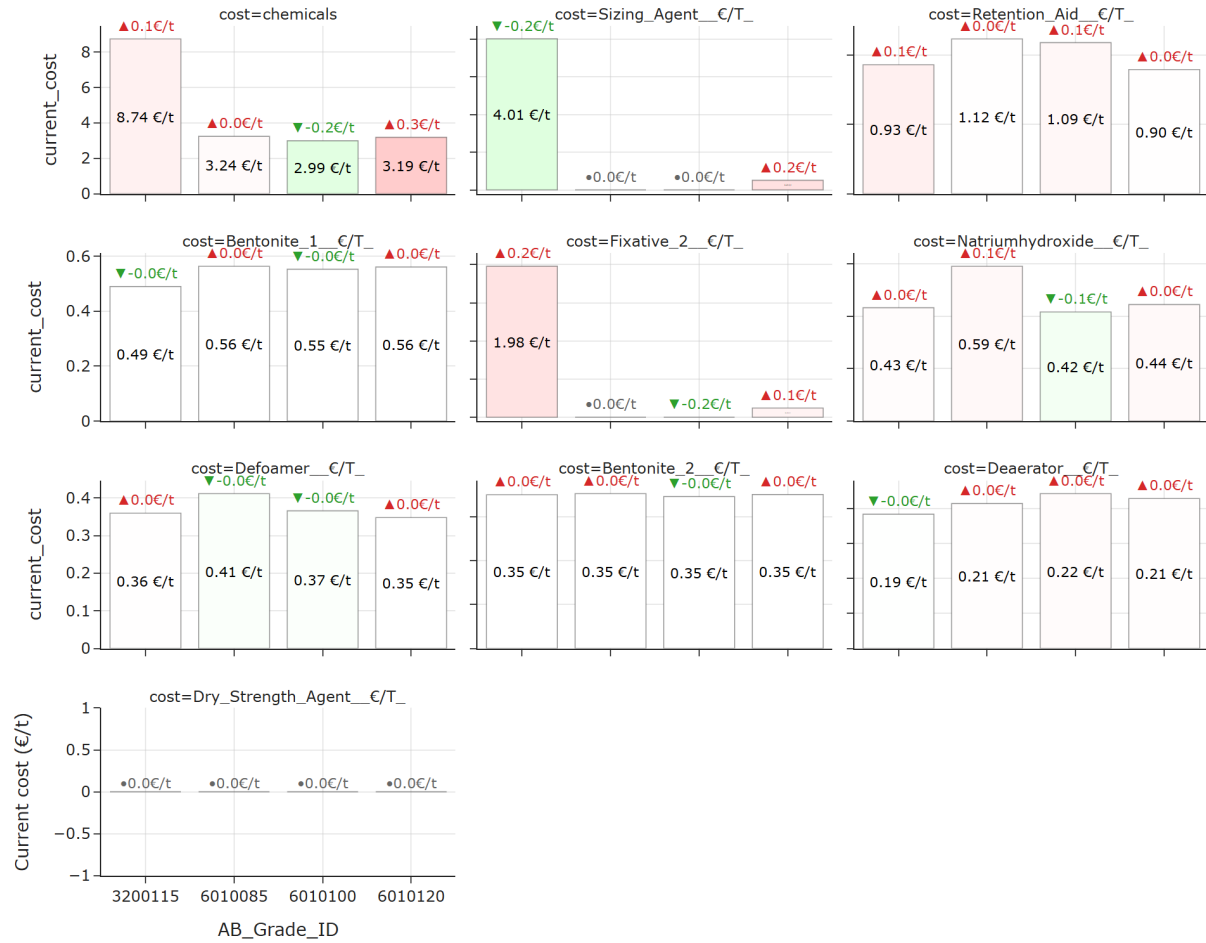


Average change by electricity:

- **Electricity_Specific** increased on average (+0.49 €/t).
- **Electricity_Wiresection** increased on average (+0.15 €/t).
- **Electricity_Freqconverters** decreased on average (-0.13 €/t).
- **Electricity_Predryer** increased on average (+0.04 €/t).
- **Electricity_Press_and_Steambox** decreased on average (-0.03 €/t).
- **Electricity_Other** increased on average (+0.01 €/t).
- **Electricity_Afterdryer** decreased on average (-0.00 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Electricity_Specific** (+0.49 €/t on average), with the highest increase in grade 6010085 (+0.92 €/t) and the largest average decrease comes from **Electricity_Freqconverters** (-0.13 €/t on

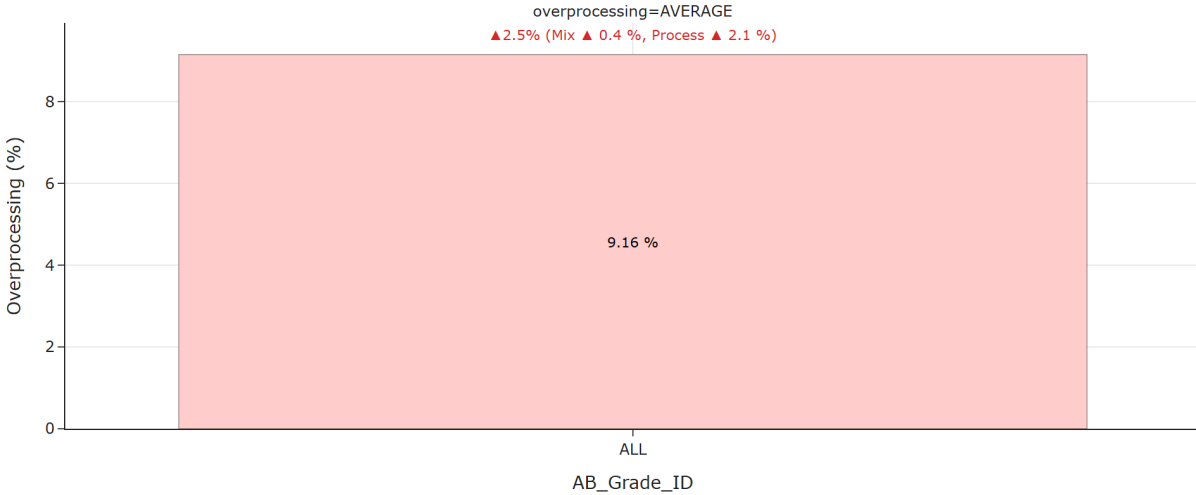
average), with the biggest drop in grade 6010120 (-0.24 €/t). These components stand out and may require further investigation.



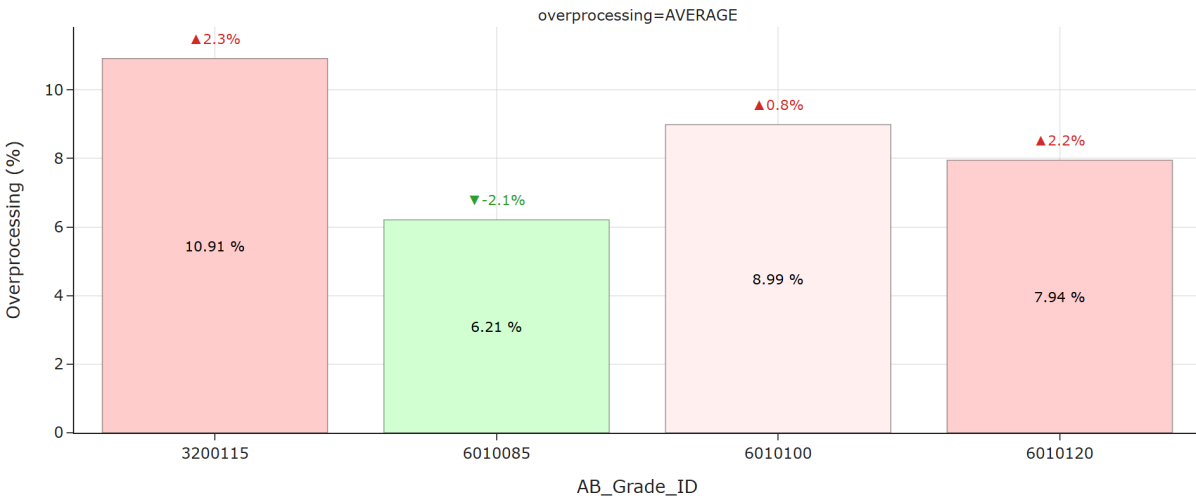
Average change by chemicals:

- **Fixative_2_€/T** increased on average (+0.14 €/t).
- **Retention_Aid_€/T** increased on average (+0.04 €/t).
- **Defoamer_€/T** decreased on average (-0.02 €/t).
- **Deaerator_€/T** increased on average (+0.01 €/t).
- **Sizing_Agent_€/T** decreased on average (-0.01 €/t).
- **Natriumhydroxide_€/T** increased on average (+0.01 €/t).
- **Bentonite_2_€/T** increased on average (+0.00 €/t).
- **Bentonite_1_€/T** decreased on average (-0.00 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Fixative_2_€/*T*** (+0.14 €/t on average), with the highest increase in grade 3200115 (+0.19 €/t) and the largest average decrease comes from **Defoamer_€/*T*** (-0.02 €/t on average), with the biggest drop in grade 6010085 (-0.04 €/t). These components stand out and may require further investigation.



For all considered grades, TOTAL overprocessing saw an increase of 2.48 %, resulting in 9.16 %.



For grade 3200115, TOTAL overprocessing saw an increase of 2.32 %, resulting in 10.91 %.

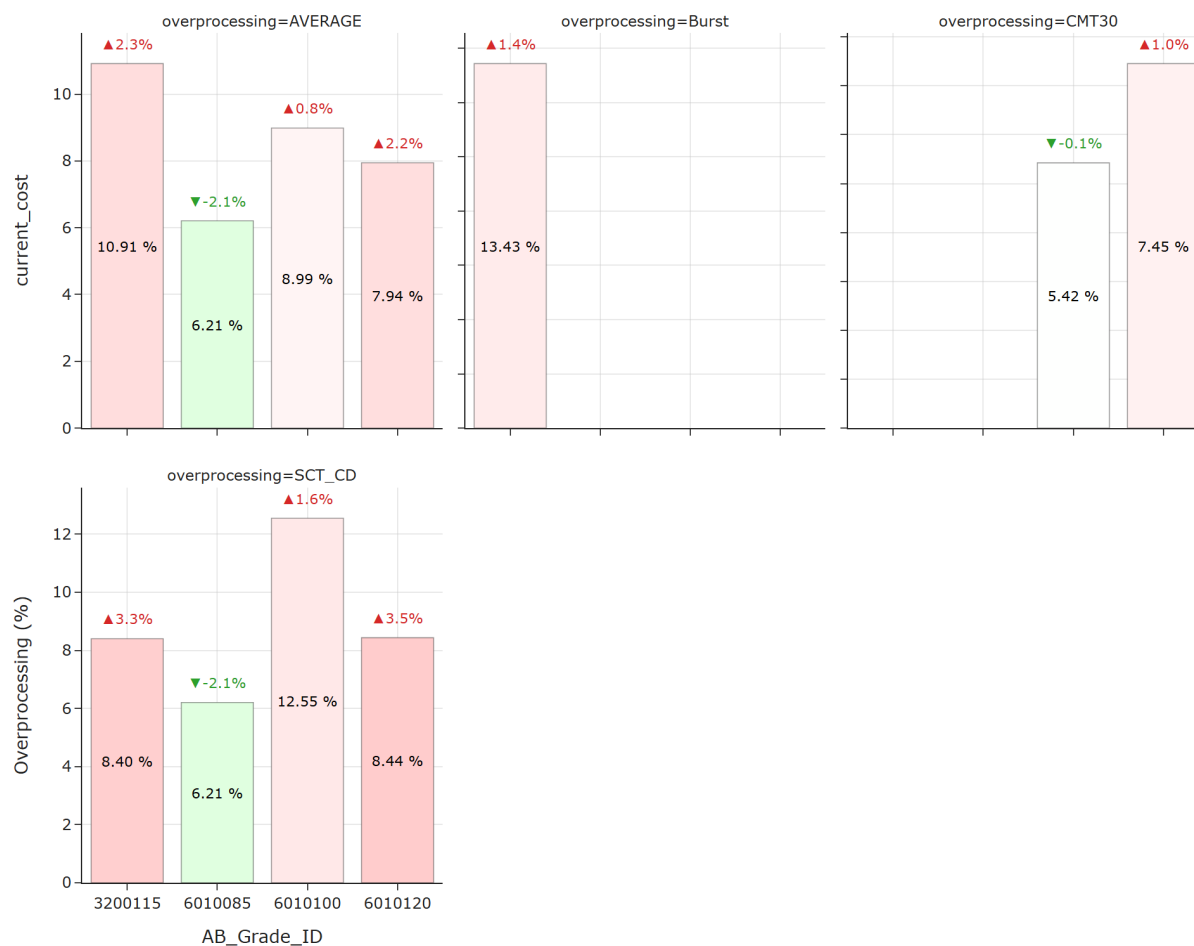
For grade 6010085, TOTAL overprocessing saw a decrease of 2.11 %, resulting in 6.21 %.

For grade 6010100, TOTAL overprocessing saw an increase of 0.75 %, resulting in 8.99 %.

For grade 6010120, TOTAL overprocessing saw an increase of 2.23 %, resulting in 7.94 %.

Looking across individual grades, the strongest overprocessing increase was observed for grade

3200115 (+2.32 %) and the strongest overprocessing decrease was observed for grade 6010085 (-2.11 %). These movements stand out and may require further investigation.



Average change by component:

- **SCT_CD** increased on average (+1.56 %).
- **Burst** increased on average (+1.37 %).
- **CMT30** increased on average (+0.44 %).

A more detailed breakdown by component shows that the largest average increase comes from **SCT_CD** (+1.56 % on average), with the highest increase in grade 6010120 (+3.48 %). These components stand out and may require further investigation.

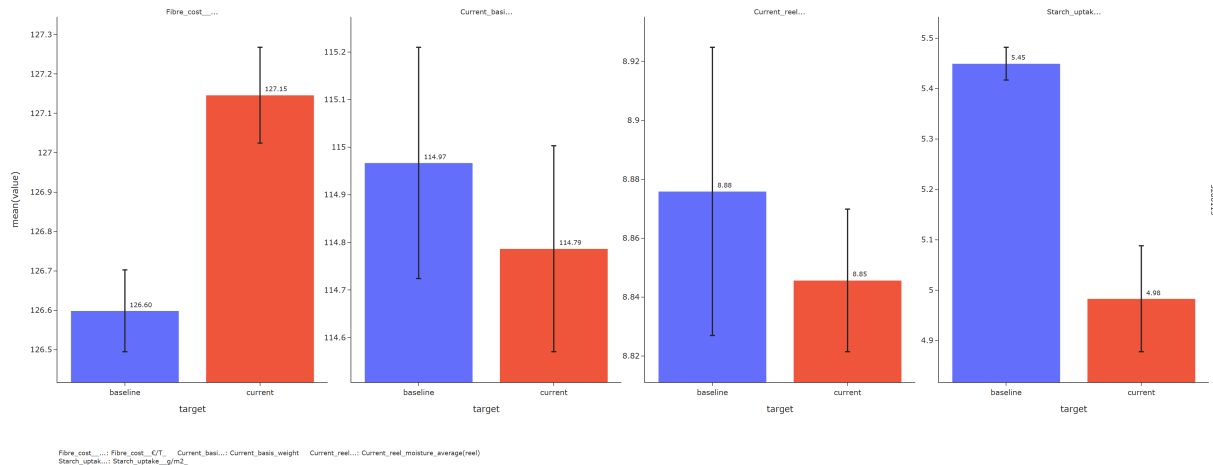
3 Breakdown by grades

3.1 3200115

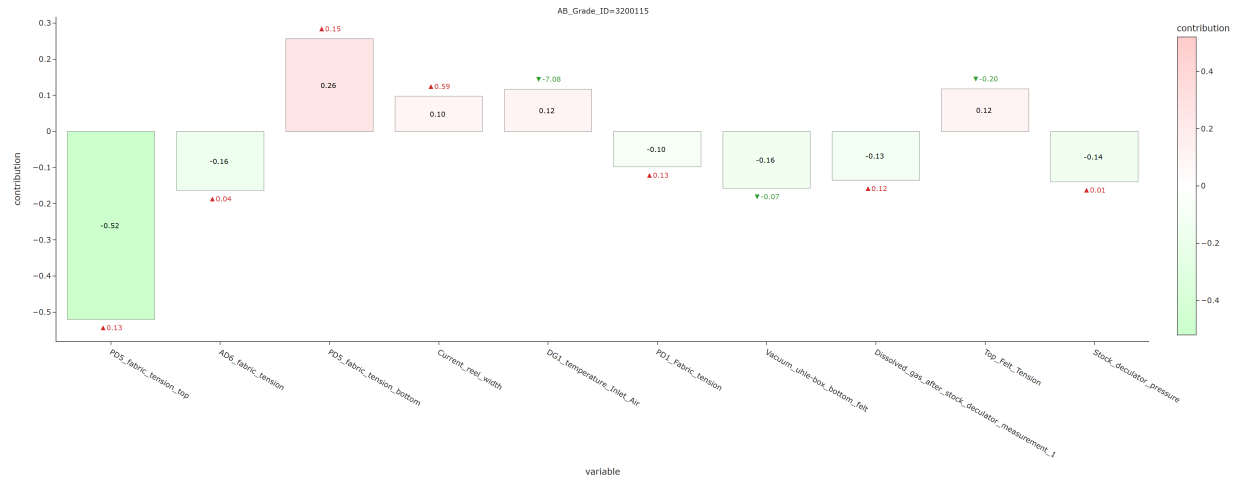
3.1.1 Fiber cost



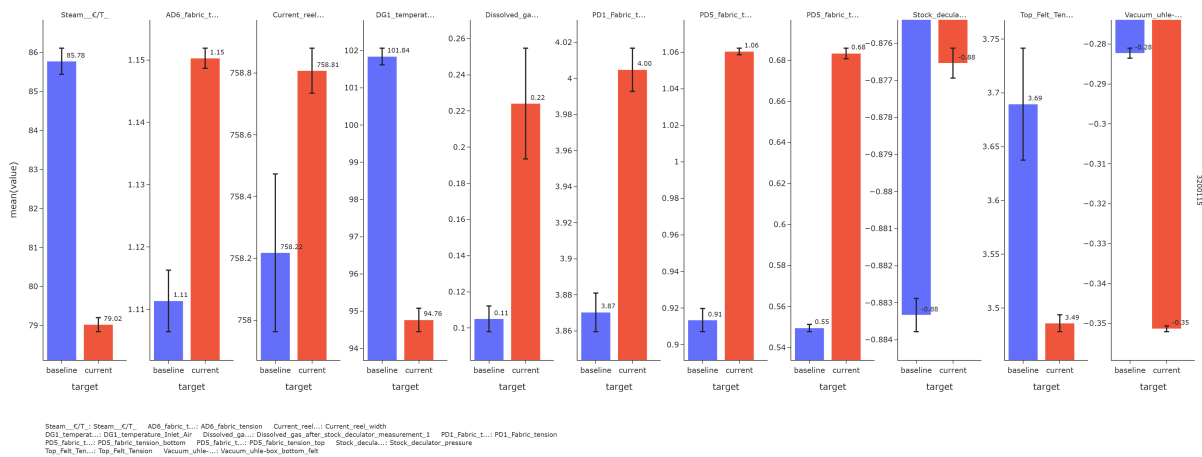
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-increasing drivers:** - Starch uptake g/m2 decreased (-0.47)



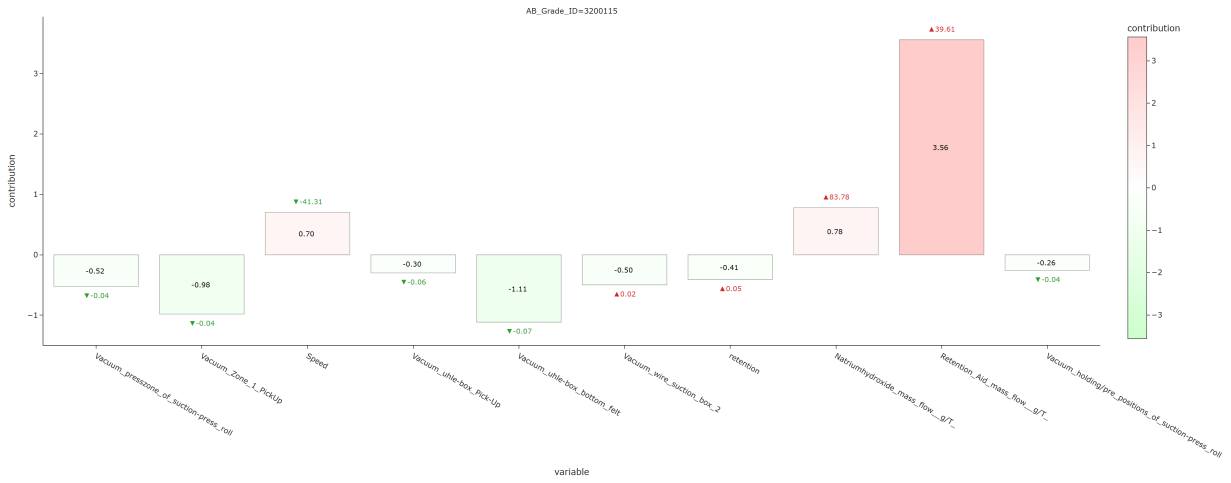
3.1.2 Steam cost



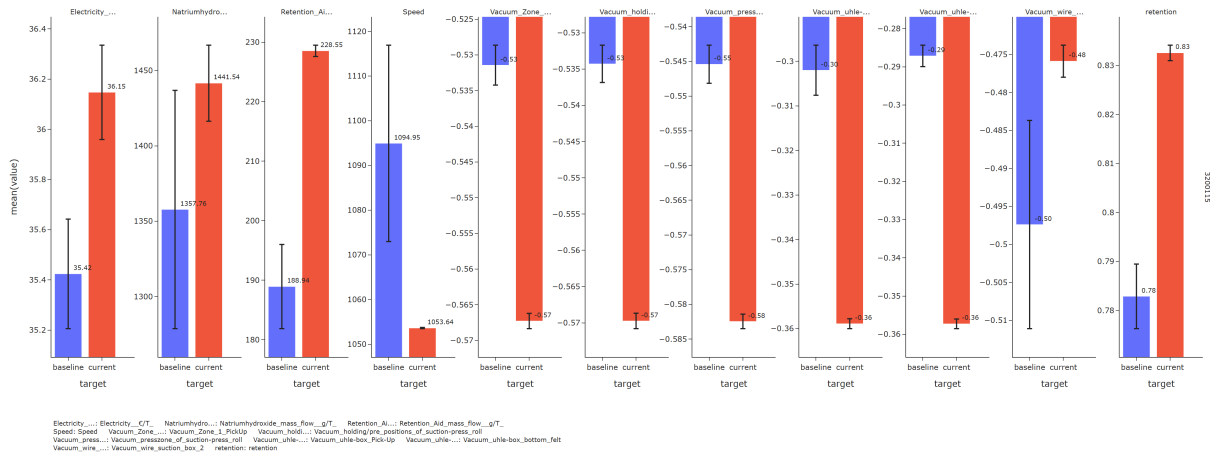
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - PD5 fabric tension top increased (+0.13) - AD6 fabric tension increased (+0.04) - Vacuum uhle-box bottom felt decreased (-0.07) - Stock decuator pressure increased (+0.01) - Dissolved gas after stock decuator measurement 1 increased (+0.12) **Cost-increasing drivers:** - PD5 fabric tension bottom increased (+0.15)



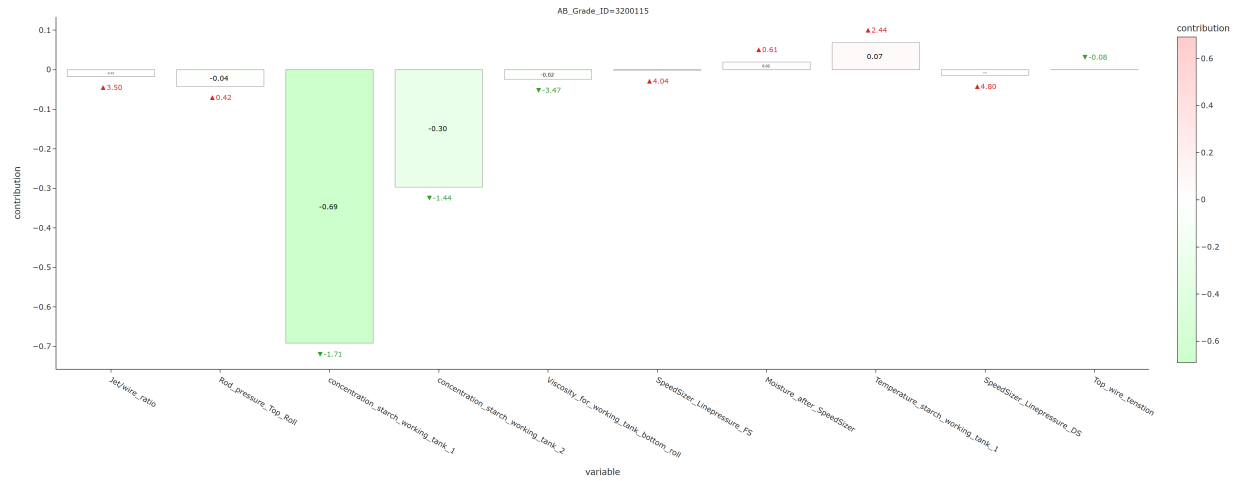
3.1.3 Electricity cost



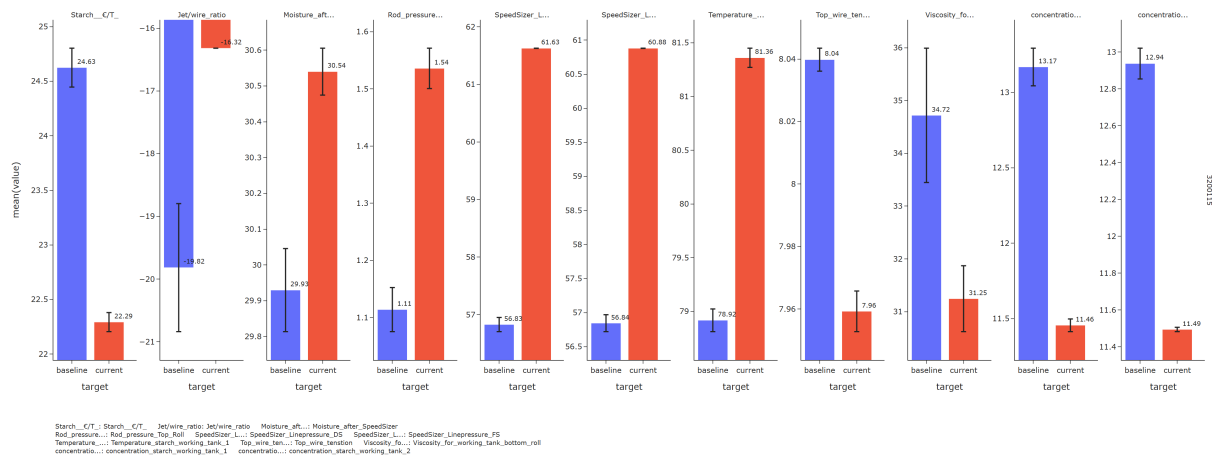
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Vacuum uhle-box bottom felt decreased (-0.07) - Vacuum Zone 1 PickUp decreased (-0.04) **Cost-increasing drivers:** - Retention Aid mass flow g/T increased (+39.61)



3.1.4 Starch cost

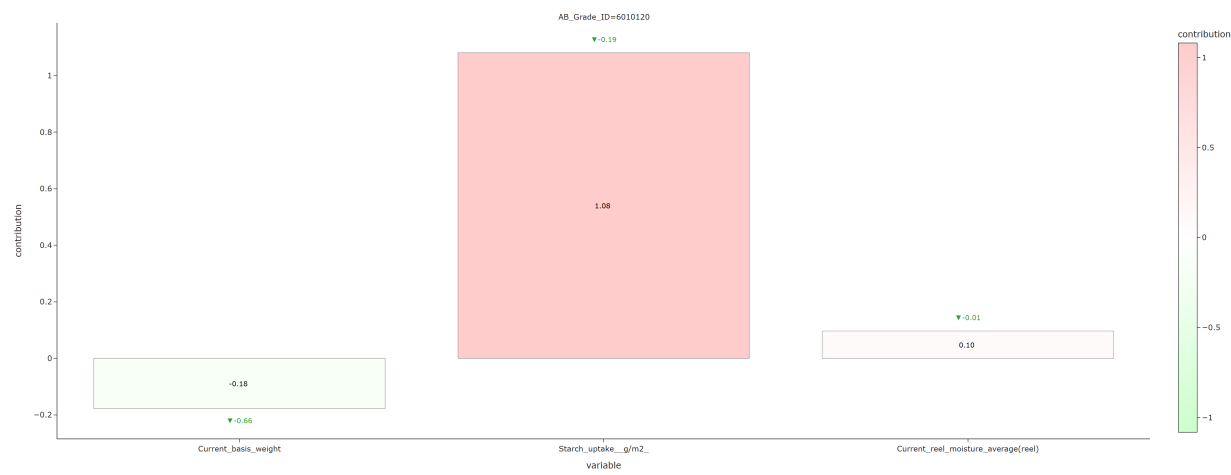


For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Concentration starch working tank 1 decreased (-1.71)
- Concentration starch working tank 2 decreased (-1.44)

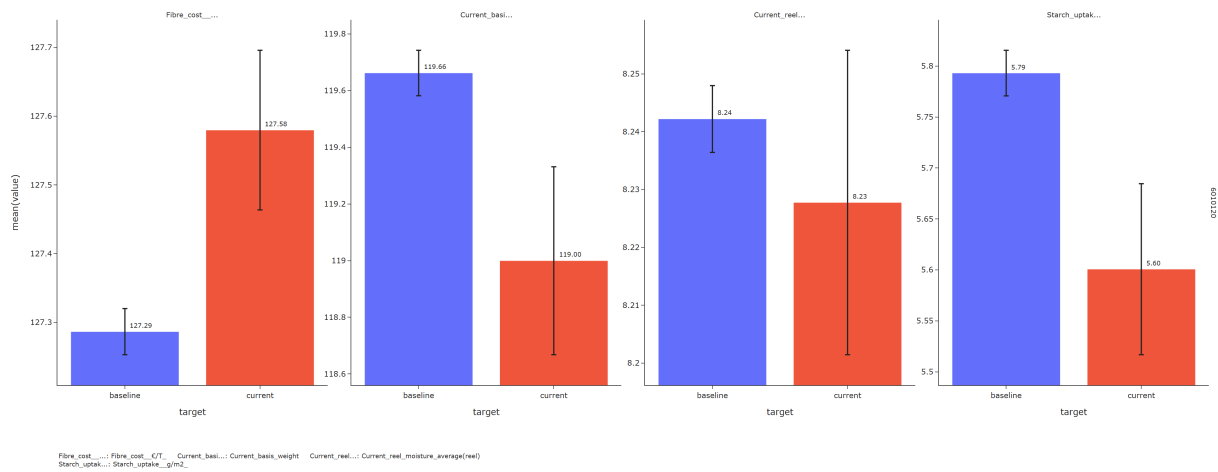


3.2 6010120

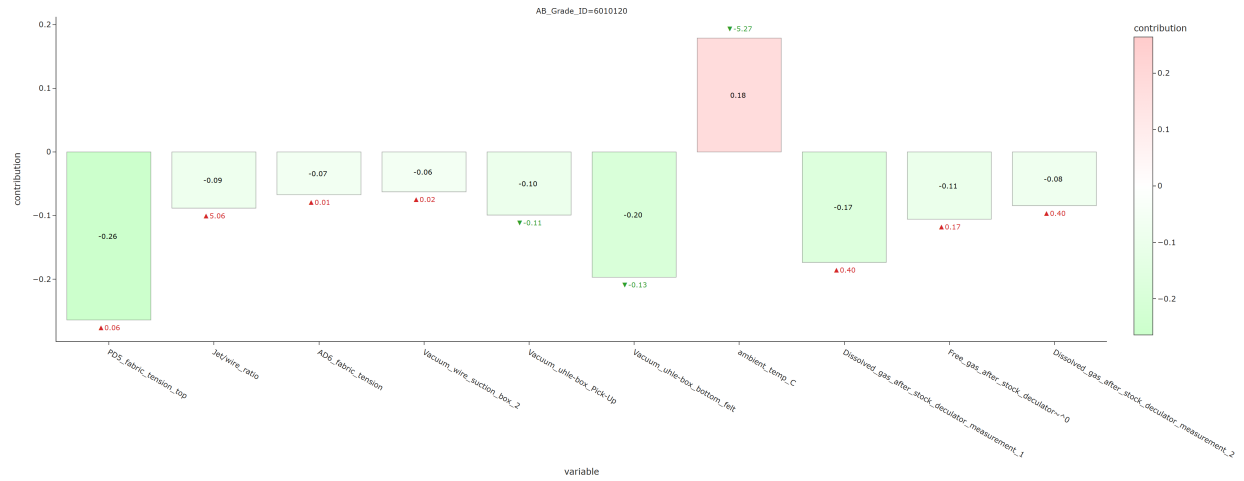
3.2.1 Fiber cost



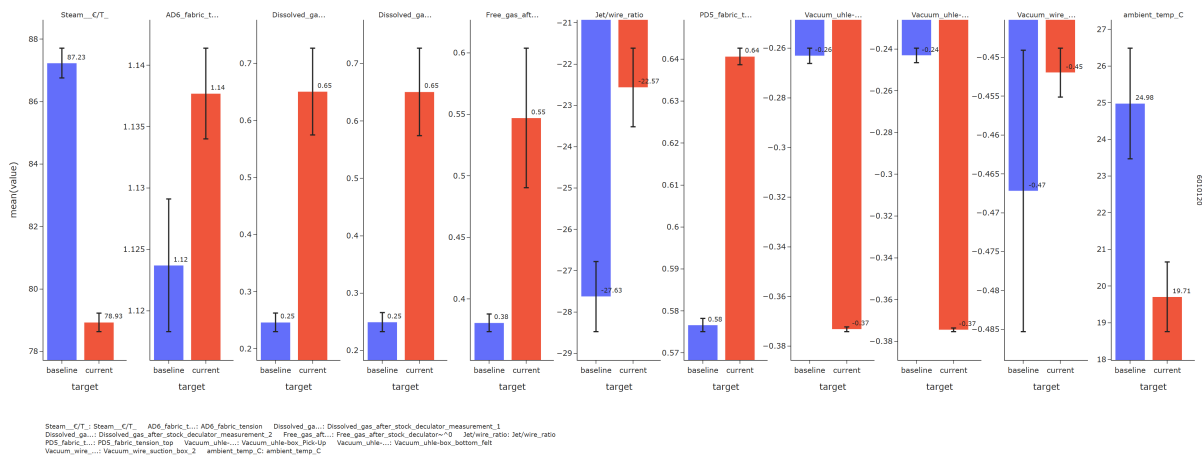
For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-increasing drivers:** - Starch uptake g/m2 decreased (-0.19)



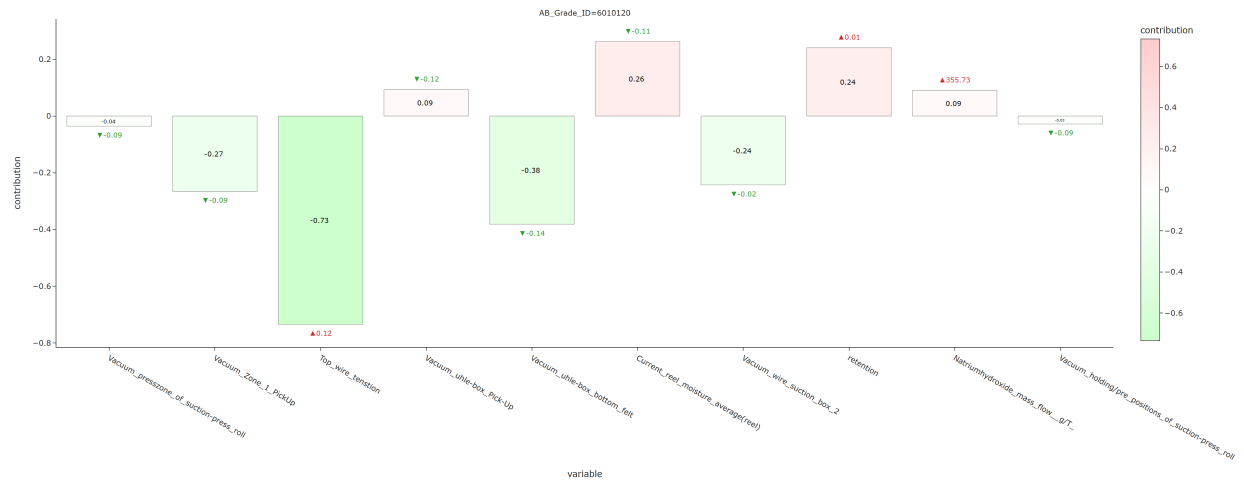
3.2.2 Steam cost



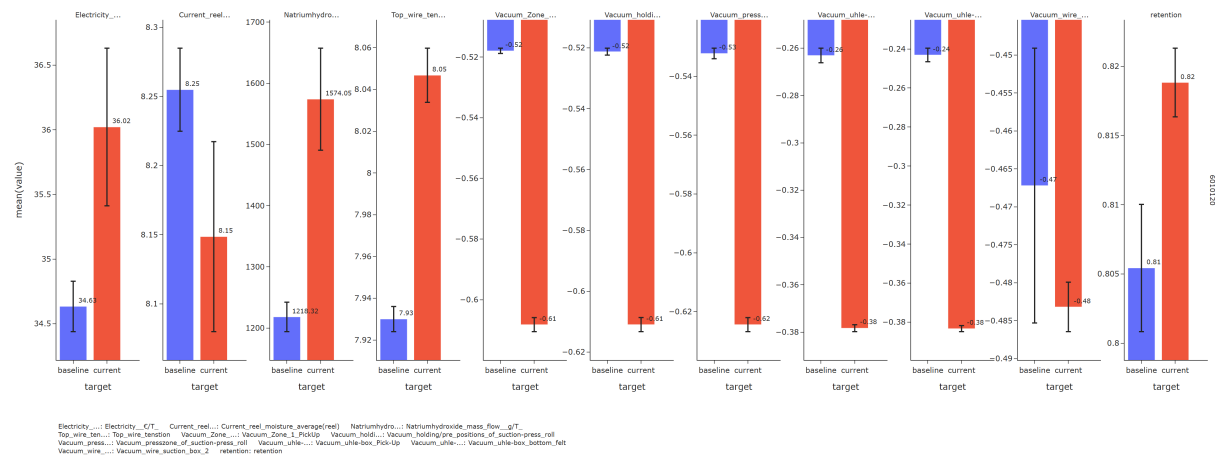
For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - PD5 fabric tension top increased (+0.06) - Vacuum uhle-box bottom felt decreased (-0.13) - Dissolved gas after stock deculator measurement 1 increased (+0.40) - Free gas after stock deculator~0 increased (+0.17) **Cost-increasing drivers:** - Ambient temp C decreased (-5.27)



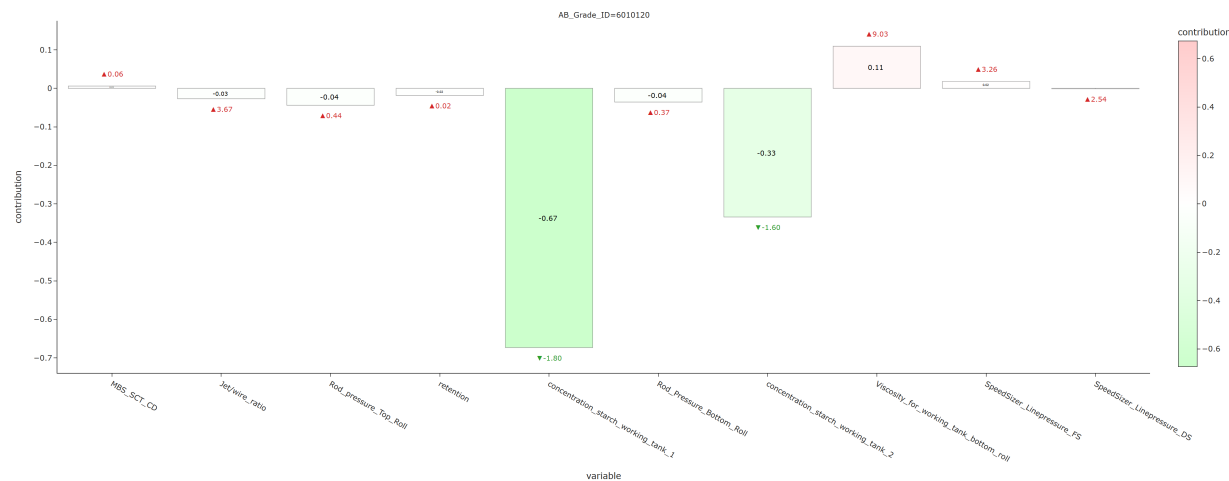
3.2.3 Electricity cost



For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Top wire tension increased (+0.12) - Vacuum uhle-box bottom felt decreased (-0.38)



3.2.4 Starch cost



For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers:** - Concentration starch working tank 1 decreased (-1.80)
- Concentration starch working tank 2 decreased (-1.60)

